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Roll # FA23-BCE-004

(Q.no.1)

$$X(n) = \{1, 2, 3, 4, 4, 3, 2, 1\}$$

 $X(k)$ using DIT-FFT

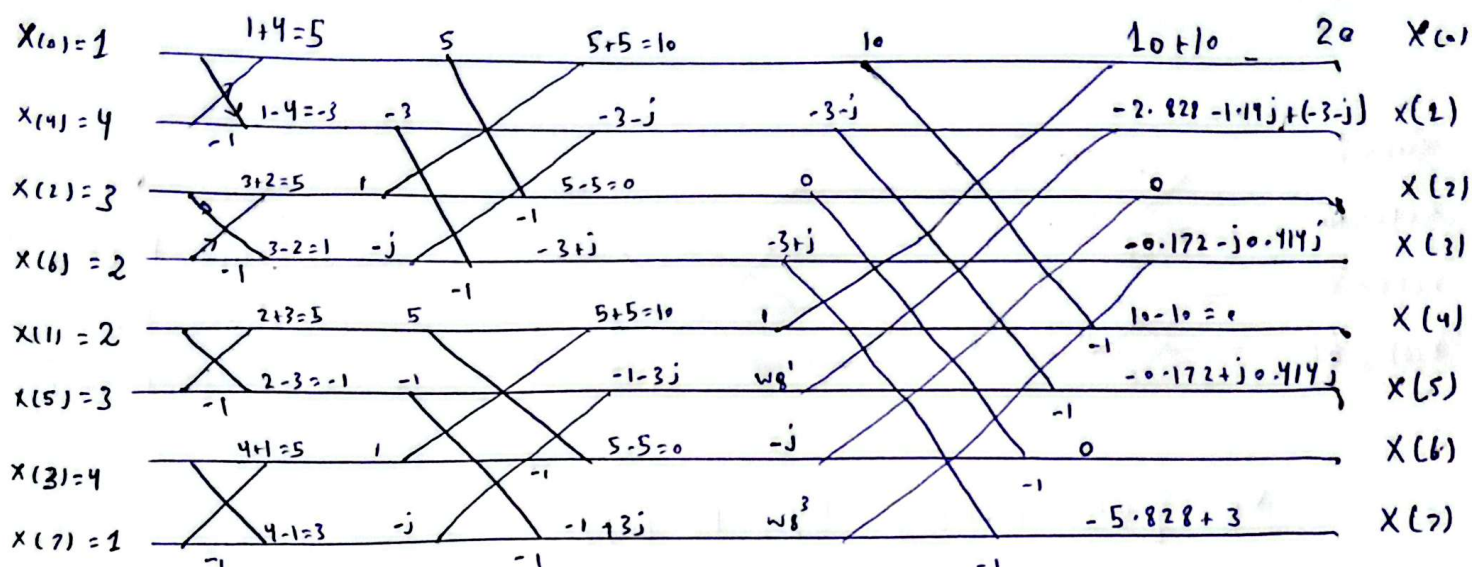
After bit reversal we get

000	000	$X(0)$	= 1	} even
001	100	$X(4)$	= 4	
010	010	$X(2)$	= 3	
011	110	$X(6)$	= 2	
100	001	$X(1)$	= 2	} odd
101	101	$X(5)$	= 3	
110	011	$X(3)$	= 4	
111	111	$X(7)$	= 1	

$$X_e(n) = \{1, 3, 4, 2\} \quad X_o(n) = \{2, 4, 3, 1\}$$

Four point DFT of even

and odd on next page



$$X(k) = \{ 20, -5.828 - 7.414j, 0, -0.172 - j0.414, 0, -0.172 + j0.414, 0, -5.828 + j7.414 \}$$

(Q.no.2)

$$x(n) = 2^n, \quad N = 8$$

Find $x(k)$

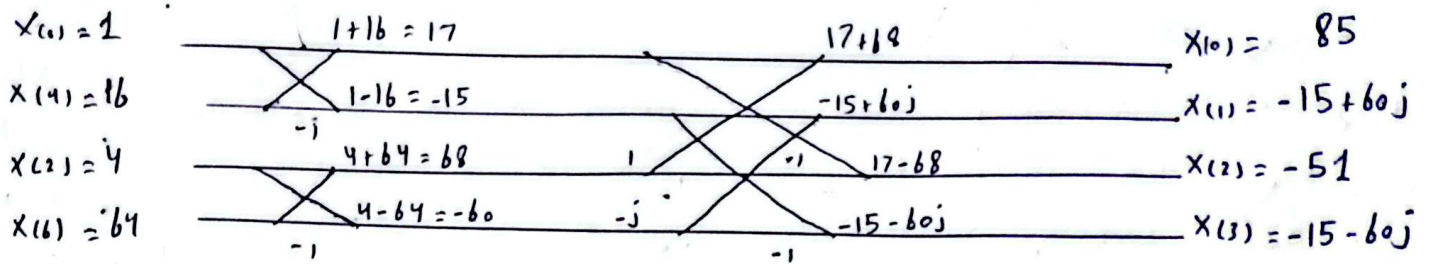
$$x(n) = \{ 1, 2, 4, 8, 16, 32, 64, 128 \}$$

Bit reversal to split into even and odd

$$x_e(k) = \{ 1, 4, 16, 64 \}$$

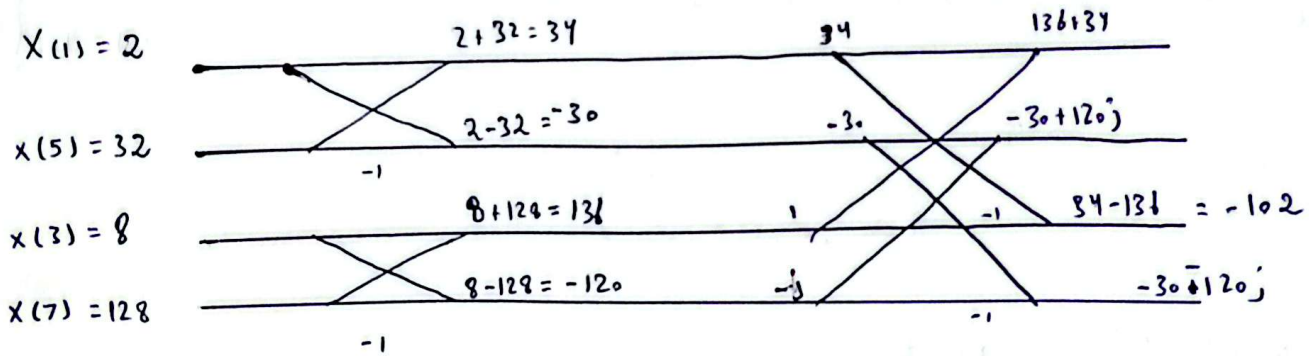
$$x_o(k) = \{ 2, 8, 32, 128 \}$$

Even:



$$X_e = \{85, -15+60j, -51, -15-60j\}$$

Odd:



$$X_o = \{170, -30+120j, -102, -30-120j\}$$

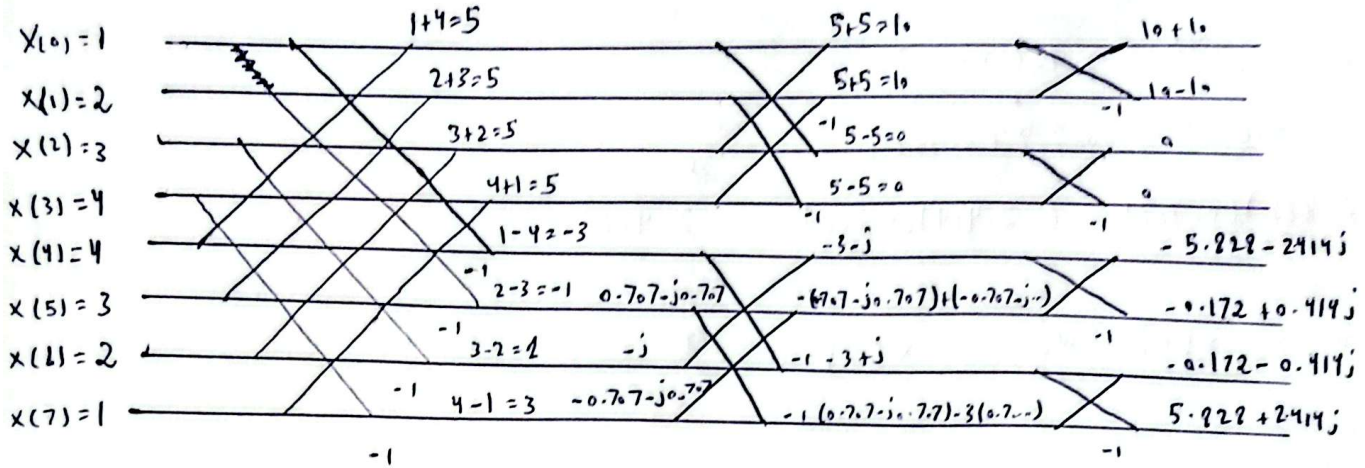
$$X(k) = X_e(k) + W_8^k X_o(k)$$

$$X(k) = \{255, -21.21 + j147.49, -51, -8.79 - j27.49, -85, -8.79 + j27.49, -51, -21.21 - j147.49\}$$

(Q.no.3)

$$X(n) = \{1, 2, 3, 4, 4, 3, 2, 1\}$$

use DIF - FFT



$$X(k) = \{20, -5.828 - 2.414j, 0, -0.172 - 0.414j, 0, -0.172 + 0.414j, 0, 5.828 + 2.414j\}$$

(Q.no.4)

$$X(k) = \{20, -5.828 - j2.414, 0, -0.172 - 0.414j, 0, -0.172 + 0.414j, 0, -5.828 + j2.414\}$$

Find $x(n)$

$$\text{Sol: } x(n) = \frac{1}{8} \sum_k x[k] e^{j \frac{2\pi n k}{8}}$$

$$k=1 \quad n=0$$

$$= \frac{1}{8} [20 [-5.828 - 2.414j] + 0.172 - 0.414j] + [(0.172 + 0.414j) + (-5.828 + 2.414j)]$$

$$= \frac{1}{8} (8) = 1$$

$$n=1$$

$$k=1 \rightarrow e^{j\pi/4} = 0.707 + 0.707j$$

$$k=3 \rightarrow e^{j3\pi/4} = -0.707 + 0.707j$$

$$= (-5.828)(0.707) - (-2.414)(0.707) = -2.414$$

$$\text{For } k=3 \quad x[3] = w_8^3 \quad \text{or } x[3] = e^{j3\pi/4}$$

$$n=4$$

$$e^{j\pi} = -1$$

$$e^{j3\pi} = -1$$

$$k=2 = 5.828$$

$$k=7 = 5.828$$

$$k=3 = 0.171$$

$$k=5 = 0.171$$

$$20 + 5.828 + 5.828 + 0.171 + 0.171 = \frac{32}{8} = 4$$

$$n=5, 6, 7$$

$$x[5] = x[3] = 3$$

$$x[6] = x[2] = 2$$

$$x[7] = x[1] = 1$$

$$x(n) = \{1, 2, 3, 4, 4, 3, 2, 1\}$$